

Dr. Christopher J. Blais

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Summary

Postdoctoral Researcher with 4 years of experience in industrial automation. Expertise in computational modelling of complex systems, optimization, and machine learning. Interested in solving complex problems by marrying simulations to the real world.

Education

Northeastern University

Ph.D. in Chemical Engineering, October 2024. GPA: 3.5

University of New Hampshire:

B.S. in chemical engineering (Honors), May 2015. GPA: 3.76

Professional Experience:

*Oak Ridge National Laboratory, **Postdoctoral Researcher***

Jan 2025-Present

Implementing numerical/linear algebra approaches to structurally categorize process data and optimize sensor network design. Using Gaussian Process Regression for setpoint optimization in a continuous industrial process.

*Chevron, **Kinetic Modelling Intern***

Dec 2022-Mar 2023

Used Python, Perl, C++ and VBA to construct a deactivation model for a naphtha reforming unit. Generated Python wrappers for legacy code that can run across multiple cpu cores.

*Northeastern University, **PhD Researcher***

June 2019- December 2024

Created and optimized microkinetic models for heterogenous catalysis. Contributed to and developed RMG source code for better prediction of catalytic systems. Performed statistical analysis and Bayesian Parameter Estimation for RMG generated models.

*Superior Controls Inc. (acquired by Etech Global Services LLC), **Project Engineer***

May 2015-June 2019

Worked on the design and scale-up of the automation systems for customers in pharmaceutical, food and beverage industries. Provided automation support and technical assistance for process areas undergoing validation.

Skills:

Software: Python, Git, Julia, VBA, MPI, OpenMP, CUDA, OpenACC, Numba, CuPy, Opencode, Ollama, Gitlab DUO, Github copilot, FactoryTalk, Studio5000, LabView, SQL.

Science: Microkinetic modelling, modelling software toolkits (Cantera, Chemkin, Ansys, Comsol), industrial sensors and communication protocols.

Mathematics: Gaussian Process Regression (GPR), Bayesian Parameter Estimation (BPE), Markov Chain Monte Carlo (MCMC), Ensemble Slice Sampling (ESS), Design of Experiments (DOE).

Extracurriculars:

Volunteering: Great Bay Estuary, Haley House South End (Boston, MA), Habitat for Humanity

Hobbies: Hiking, camping, distance running, Godot for video game development, cooking.

Conference Presentations

“Exploring the Catalytic Combustion Mechanism of Hydroxylammonium Nitrate Using RMG-Cat” AIChE Annual Meeting. Online. November 16 – November 20 2020.

“Using RMG to investigate the Catalytic Combustion of Hydroxylammonium Nitrate” 39th Northeast Regional Meeting on Kinetics and Dynamics. Online. January 30 2021.

“Using Reaction Mechanism Generator to Investigate the Catalytic Synthesis of MeOH” AIChE Annual Meeting. Boston, MA. November 16 – November 20 2021.

“Catalytic Methanol Synthesis Mechanism Development Using the Reaction Mechanism Generator” 27th North American Catalysis Society Meeting. Manhattan, NY. May 22 – May 27 2022.

“Observability and Redundancy Labelling Methodology in Pressurized Water Treatment Systems” AIChE Annual meeting. Boston, MA. November 2 – November 6 2025.

Publications

Blais C., “Microkinetic Modelling and Uncertainty Quantification on Differing Crystal Facets” Northeastern University ProQuest Dissertations & Theses, 2024. 31636639.

Blais, C., Xu, C., West, R. H., “Analyzing the Uncertainty of Linear Scaling, Machine Learning, and DFT Derived Thermodynamics for the Catalytic Partial Oxidation of Methane on Rhodium 111”, *Journal of Physical Chemistry C*, 2024, 128 (41) 17418-17433

Blais, C. Mukherjee, S. Ghorbani, P. Ghanem, S., Rosso, D., Villez, K., (In Review), “Accurate Evaluation of Observability and Redundancy in Water Production Systems” (Submitted June, 2026).